Current Hybrid Architecture Performance
Second to Sixth Drawing Sets | Version 10

MODEL_VERSION: 10.11.24
GATE: P2610E3_SECOND_TO_SIXTH_FULL_POPULATION_LIVE_VISION_HYBRID_ACCURACY_BENCHMARK_AND_RE
PHASE: P2.6.10-E.3
DECISION: PASS_WITH_LIMITATIONS
BENCHMARK MODE: LIVE_BENCHMARK
Confidential / internal use. Shadow benchmark. Not a production promotion.

1. EXECUTIVE SUMMARY
Five-set current hybrid architecture: Claude Vision semantics + deterministic engineering.
Model beams: 530 GT beams: 605 Matched: 515
HYBRID: 486 (91.70%) FALLBACK: 44 (8.30%)
Pooled beam identification: 85.12% (515.0 / 605.0)
Pooled bar identification: 48.17% (2008.0 / 4169.0)
Pooled correct-of-detected: 36.50% (733.0 / 2008.0)
Pooled diameter identification: 77.84%
Pooled steel/weight accuracy: 60.64%
Pooled overall accuracy: 57.61%
Model kg: 90305.121 Benchmark kg: 148910.644

2. CURRENT HYBRID ARCHITECTURE
DXF -> Beam discovery -> Deterministic geometry context -> Claude Vision
-> D.1 semantic authority contract -> D.2 hybrid resolution
-> D.3 engineering binding -> D.4 shadow engineering calculation
-> accuracy benchmark.
VISION SEMANTICS: target identity, layer, groups, bar count, diameter,
specification, MAIN/EXTRA, support scope, stirrup identification.
DETERMINISTIC ENGINEERING: spacers, geometry, cut length, development,
anchorage, hooks, stirrup engineering, pieces, weight, BBS, workbook.
Vision-preferred is validated, not blindly accepted.

3. BENCHMARK SCOPE AND POPULATION
Second, Third, Fourth, Fifth, Sixth. First Set excluded.
Second: model=65 GT=67 matched=64 truth=ESTIMATOR_EXCEL
Third: model=61 GT=63 matched=57 truth=ESTIMATOR_EXCEL
Fourth: model=118 GT=143 matched=112 truth=ESTIMATOR_EXCEL
Fifth: model=143 GT=187 matched=143 truth=ESTIMATOR_EXCEL
Sixth: model=143 GT=145 matched=139 truth=ESTIMATOR_EXCEL

4. VISION EXECUTION AND HYBRID COVERAGE
Second: eligible=63 attempted=63 new=63 reused=0 retried=0 api=63 schema=63 usable=63 HYBRID=63 FALLBACK=2
Third: eligible=61 attempted=61 new=61 reused=0 retried=0 api=61 schema=61 usable=61 HYBRID=61 FALLBACK=0
Fourth: eligible=76 attempted=76 new=76 reused=0 retried=0 api=76 schema=76 usable=76 HYBRID=76 FALLBACK=42

Sixth: eligible=143 attempted=143 new=143 reused=0 retried=0 api=143 schema=143 usable=143 HYBRID=143 FALLBACK=0
Fifth Set reuse decision: VISION_REUSED_CURRENT_ARCHITECTURE

5. ACCURACY BY DRAWING SET
Second: beam=95.52% bar=65.28% correct=31.35% diameter=78.97% steel=87.95% overall=70.03%
Third: beam=90.48% bar=54.29% correct=27.27% diameter=74.31% steel=68.25% overall=60.07%
Fourth: beam=78.32% bar=40.43% correct=41.32% diameter=75.26% steel=54.29% overall=53.59%
Fifth: beam=76.47% bar=39.29% correct=36.89% diameter=76.75% steel=47.58% overall=50.06%
Sixth: beam=95.86% bar=59.83% correct=39.38% diameter=81.85% steel=83.18% overall=69.56%

6. POOLED SECOND-TO-SIXTH PERFORMANCE
Headline KPIs pool raw numerators and denominators. Steel kg is pooled before accuracy.
Set percentages are not averaged for the headline overall score.
Overall = mean(beam ID, bar ID, correct-of-detected, steel/weight). Diameter excluded.

7. DIAMETER IDENTIFICATION (DETECTED BAR LINES)
QA.2A diameter_accuracy_pct is MATCH/detected and is not used here.
A detected bar is diameter-correct unless status is WRONG_DIAMETER.
GT diameter is the estimator line diameter. Pooled from raw counts. Diameter excluded from overall.
Diameter | GT bar lines | Detected | MATCH | WRONG_DIA | Diameter ID | Note
Ø8: GT=358 detected=152 MATCH=32 WRONG_DIA=3 ID=98.03% Usually right when found; many missing
Ø10: GT=714 detected=214 MATCH=56 WRONG_DIA=11 ID=94.86% Usually right when found; many missing
Ø12: GT=573 detected=207 MATCH=68 WRONG_DIA=28 ID=86.47% Usually right when found; many missing
Ø16: GT=588 detected=372 MATCH=169 WRONG_DIA=112 ID=69.89% Frequent diameter swaps
Ø20: GT=784 detected=436 MATCH=193 WRONG_DIA=155 ID=64.45% Frequent diameter swaps
Ø25: GT=1097 detected=616 MATCH=210 WRONG_DIA=130 ID=78.90%
Ø32: GT=26 detected=11 MATCH=5 WRONG_DIA=6 ID=45.45% Low volume - percentage unstable
TOTAL scored Ø8-Ø32: GT=4140 detected=2008 MATCH=733 WRONG_DIA=445 ID=77.84% Pooled headline from raw counts

8. DIAMETER-WISE STEEL QUANTITY (NOT THE SAME AS DIAMETER IDENTIFICATION)
Quantity ratio = automated kg / estimated kg x 100. It is not accuracy.
A ratio above 100% is an overestimate. kg pooled before difference and ratio.
Diameter | Estimated kg | Automated kg | Difference kg | Abs % diff | Quantity ratio
Ø8: est=6515 auto=5051 diff=-1464 abs=22.5% ratio=78%
Ø10: est=24555 auto=14163 diff=-10393 abs=42.3% ratio=58%
Ø12: est=22253 auto=12722 diff=-9531 abs=42.8% ratio=57%
Ø16: est=11598 auto=7678 diff=-3920 abs=33.8% ratio=66%
Ø20: est=35888 auto=20506 diff=-15382 abs=42.9% ratio=57%
Ø25: est=45294 auto=28282 diff=-17011 abs=37.6% ratio=62%
Ø32: est=2808 auto=1901 diff=-906 abs=32.3% ratio=68%
TOTAL: est=148911 auto=90303 diff=-58607 abs=39.4% ratio=61%

9. SEMANTIC ERROR PROFILE
MISSING: 2161
MATCH: 733
WRONG_QUANTITY: 593
WRONG_DIAMETER: 445
EXTRA: 183
WRONG_ROLE: 134

ACCEPTABLE_EXTRA: 89

10. ENGINEERING CALCULATION PROFILE
CUT_LENGTH_DERIVED_FALLBACK: 188
CUT_LENGTH_UNAVAILABLE: 0
STIRRUP_ENGINEERING_UNAVAILABLE: 69
PARTIAL_CALCULATION: 82
INCOMPATIBLE: 0
SPACER_ZERO: 242

11. HYBRID / FALLBACK COHORTS (pooled applicable subsets)
HYBRID_ONLY: beam=97.52% bar=57.03% correct=36.39% steel=75.22% overall=66.54%
FALLBACK_ONLY: beam=97.73% bar=48.43% correct=38.21% steel=71.00% overall=63.84%
FULL_POPULATION: beam=85.12% bar=48.17% correct=36.50% steel=60.64% overall=57.61%

12. CURRENT WORKFLOW VALUE
Model-assisted estimation. Estimator verification required.
Accuracy is not equivalent to time saved. No time study in this phase.
Do not treat Vision API success as engineering accuracy.

13. METHODOLOGY, SOURCES AND LIMITATIONS
Truth: estimator Excel (evaluation-only). QA.2A BeamMatcher / BarMatcher / metric8.
QA.3.0 four-KPI overall. Ground truth is never used in runtime semantic resolution.
Limitations: ESTIMATOR_WORKBOOK_MAPPING_LIMITATION, FIRST_SET_EXCLUDED, SHADOW_ONLY_NOT_PRODUC
PRODUCTION_WRITE=false ENGINEERING_CHANGES=NONE shadow_only=true

COST / EXECUTION
New live: 343 Reused: 143 Retried: 0 Failed: 0
Input tokens: 1088060 Output tokens: 213497 Runtime s: 3854.94

CONCLUSION
This benchmark reports current hybrid architecture performance on Second through Sixth sets. It is not a historical improvement
No historical improvement claim. No production readiness claim.